

# Early Predictors of Performance in First-Year Statistics: Attendance, Academic Integrity, and Quiz Results

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## Abstract

Sitting proximity to higher-achieving peers predicts improved exam performance among first-year statistics students. Observational and survey data from **49 students**—covering attendance, quiz scores, seating location, and indicators of dishonesty—were analysed using OLS, autoregressive, and 2SLS regression models. Students seated near higher-performing classmates scored significantly higher; the association persisted after controlling for measured dishonesty. The dishonesty index correlated strongly with lower performance, while attitude toward mathematics was a weak and insignificant predictor. Results align with prior peer-effect findings and extend them by explicitly measuring and ruling out detectable cheating as an alternative mechanism. The small sample warrants replication; if robust, the findings suggest that **seating arrangements could serve as a low-cost intervention** to boost early-course achievement.

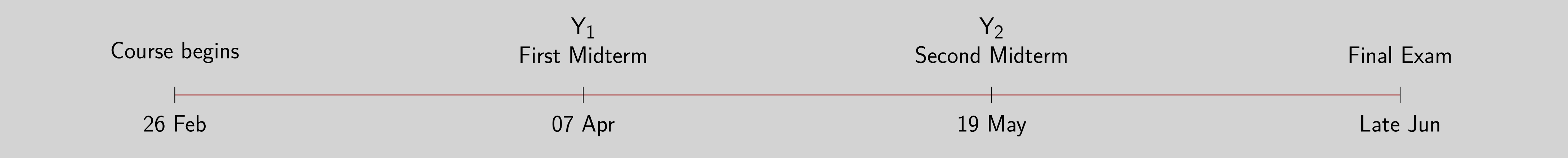
## Sample & Study Design

The final sample consisted of **49 regular (non-exam-only) students** enrolled in the mandatory course *Statistics for Economists* (4EF113421) across all five Faculty of Economics majors at the University of Goce Delchev (UGD), Shtip. Of these, 30 completed the first knowledge-assessment quiz.

To preserve natural behaviour, experimental procedures were **not disclosed** to participants. Quizzes were framed as knowledge checks with no grade impact, administered via Google Forms. Baseline variables included attendance indices, quiz participation and scores, homework completion (on-paper vs. online), late submissions, honesty checks, and a 1–6 homework-difficulty rating. Related variables were aggregated into summary indices to address the high-dimensional, hybrid time-cross-sectional structure of the data.

| Variable                        | Type         | Description                                               |
|---------------------------------|--------------|-----------------------------------------------------------|
| averATND <sub>1/2</sub>         | cont. [0, 1] | Average attendance rate (paper list, evidence list, quiz) |
| DISS <sub>1/2</sub>             | cont. [0, +] | Dishonesty index (honHW + CHEAT + honLST, weighted)       |
| numHW <sub>1/2</sub>            | cont./log    | Number of homeworks submitted                             |
| paperHW <sub>1/2</sub>          | cont./log    | Proportion submitted by hand                              |
| lateHW <sub>1/2</sub>           | cont./log    | Proportion submitted after deadline                       |
| additHW <sub>1/2</sub>          | log          | Additional (project) homework submitted                   |
| averQIZZ <sub>1/2</sub>         | cont. [0, 1] | Average in-session quiz score                             |
| hardHW_H <sub>1/2</sub>         | binary       | Above-mean perceived homework difficulty                  |
| PROXI25 <sub>1/2</sub>          | cont. [0, 1] | Proportion of sessions seated near a bottom-25% peer      |
| PROXI75 <sub>1/2</sub>          | cont. [0, 1] | Proportion of sessions seated near a top-75% peer         |
| atitMATH                        | categ. 1–6   | Self-reported attitude toward mathematics                 |
| Y <sub>1</sub> , Y <sub>2</sub> | cont. [0, 1] | First / second midterm score (out of 20 pts)              |

## Study Timeline



## Hypotheses and Functional Forms

**H0** Attendance, homework completion/timeliness, and academic honesty predict test performance **more strongly** than attitude toward mathematics:

$$Y_{1/2} = \alpha + \beta_1 \cdot \text{GENDER} + \beta_2 \cdot Y_{1/2}\text{GROUP} + \beta_3 \cdot \text{averATND}_{1/2} + \beta_4 \cdot \text{numHW}_{1/2} + \beta_5 \cdot \text{additHW}_{1/2} + \beta_6 \cdot \text{DISS}_{1/2} + \beta_7 \cdot \text{atitMATH}$$

**H1** Seating proximity to higher-achieving peers improves performance relative to students seated near the median:

$$Y_{1/2} = \alpha + \beta_1 \cdot \text{GENDER} + \beta_2 \cdot Y_{1/2}\text{GROUP} + \beta_3 \cdot \text{PROXI25}_{1/2} + \beta_4 \cdot \text{PROXI75}_{1/2}$$

**H2** Lower attendance, late homework, digitally completed homework, low quiz scores, and perceived homework difficulty predict lower final results:

$$Y_{1/2} = \alpha + \beta_1 \cdot \text{GENDER} + \beta_2 \cdot Y_{1/2}\text{GROUP} + \beta_3 \cdot \text{averATND\_H}_{1/2} + \beta_4 \cdot \text{lateHW}_{1/2} + \beta_5 \cdot \text{paperHW}_{1/2} + \beta_6 \cdot \text{averQIZZ\_H}_{1/2} + \beta_7 \cdot \text{hardHW\_H}_{1/2}$$

## DISS — Opportunistic Dishonesty Index

Constructed to measure student attempts to misrepresent their engagement or performance to the instructor. Three components are weighted by severity:

- ▶ **honHW > 1 (+1 pt)**: falsely claiming homework was completed when it was not, detected by comparing quiz self-reports against actual submissions.
- ▶ **CHEAT > 1 (+2 pts)**: observed behaviour indicative of possible cheating during midterms, not conclusive enough to trigger formal disciplinary sanctions (automatic fail + one-year ban).
- ▶ **honLST > 1 (+3 pts)**: falsifying attendance by signing the evidence list while absent, detected by auditing the list one session after each lecture.

$$\text{DISS} = 1 \cdot \text{honHW} + 2 \cdot \text{CHEAT} + 3 \cdot \text{honLST}$$

*Note:* The Y<sub>2</sub> index accumulates from Y<sub>1</sub>—behavioural corrections made between the two midterm periods do not reduce the score.

## Predicting Prior Knowledge (firstQIZZ)

OLS on demographic variables alone ( $N = 25$ ) yields no significant predictors. Key omitted variable: prior mathematics grade (prerequisite course).

| Variable                                           | Coef.   | Std. Err.        | <i>t</i> | <i>p</i> |
|----------------------------------------------------|---------|------------------|----------|----------|
| C                                                  | 0.2594  | 0.1216           | 2.133    | 0.0461   |
| ANONINDX                                           | 0.0029  | 0.0024           | 1.175    | 0.2546   |
| GENDER                                             | 0.0981  | 0.0707           | 1.388    | 0.1812   |
| HSCH                                               | 0.0053  | 0.0423           | 0.125    | 0.9015   |
| MJR                                                | -0.0429 | 0.0317           | -1.353   | 0.1919   |
| atitMATH                                           | -0.0056 | 0.0226           | -0.250   | 0.8054   |
| <i>R</i> <sup>2</sup> / Adj. <i>R</i> <sup>2</sup> | 0.2027  | / −0.0071        |          |          |
| <i>F</i> -stat                                     | 0.966   | <i>p</i> = 0.463 |          |          |

- ▶ **ANONINDX** — anonymised index (control)
- ▶ **GENDER** — binary (0/1)
- ▶ **HSCH** — 1 = general, 2 = econ./law, 3 = other
- ▶ **MJR** — enrolled major (categorical)
- ▶ **atitMATH** — attitude toward maths (1–6)

## Key Findings by Hypothesis

| Coef. (Std. Err.); stars on coef. denote significance. * $p < 0.10$ , ** $p < 0.05$ , *** $p < 0.01$ |                      |                     |                     |                     |                     |                      |                    |                     |                      |                    |
|------------------------------------------------------------------------------------------------------|----------------------|---------------------|---------------------|---------------------|---------------------|----------------------|--------------------|---------------------|----------------------|--------------------|
| $N$                                                                                                  | 1h0-r<br>27          | 2h0<br>27           | 1h1<br>46           | 1h1-d<br>46         | 1h1-i<br>46         | 2h1<br>44            | 2h1-d<br>44        | 2h1-i<br>44         | 1h2-r<br>46          | 2h2-r<br>29        |
| C                                                                                                    | 0.912***<br>(0.189)  | 0.120<br>(0.358)    | 0.455***<br>(0.121) | 0.481***<br>(0.130) | 0.418**<br>(0.158)  | 0.116<br>(0.122)     | 0.285**<br>(0.128) | 0.019<br>(0.171)    | 0.563***<br>(0.068)  | 0.094<br>(0.182)   |
| GENDER                                                                                               | 0.097<br>(0.058)     | 0.144<br>(0.084)    | 0.095<br>(0.062)    | 0.092<br>(0.063)    | 0.098<br>(0.064)    | 0.143*<br>(0.074)    | 0.122*<br>(0.069)  | 0.138**<br>(0.066)  | 0.143***<br>(0.041)  | 0.275**<br>(0.105) |
| $Y_{1/2}$ GRP                                                                                        | −0.147**<br>(0.053)  | 0.056<br>(0.076)    | −0.108**<br>(0.041) | −0.110**<br>(0.041) | −0.100**<br>(0.044) | 0.104<br>(0.069)     | 0.079<br>(0.064)   | 0.070<br>(0.061)    | −0.112***<br>(0.040) | 0.095<br>(0.096)   |
| AVERATND                                                                                             | −0.090*<br>(0.045)   | −0.087<br>(0.075)   |                     |                     |                     |                      |                    |                     | −0.131**<br>(0.050)  | 0.088<br>(0.143)   |
| NUMHW                                                                                                | 0.003<br>(0.038)     | 0.247**<br>(0.094)  |                     |                     |                     |                      |                    |                     |                      |                    |
| PROXI25                                                                                              |                      |                     | 0.004<br>(0.095)    | 0.005<br>(0.096)    | 0.005<br>(0.096)    | −0.095<br>(0.094)    | −0.083<br>(0.087)  | −0.028<br>(0.086)   |                      |                    |
| PROXI75                                                                                              |                      |                     | 0.142**<br>(0.065)  | 0.140**<br>(0.066)  | 0.211*<br>(0.119)   | 0.128<br>(0.084)     | 0.160**<br>(0.078) | 0.488***<br>(0.166) |                      |                    |
| P75×DISS                                                                                             |                      |                     | −0.039<br>(0.054)   |                     |                     | −0.132**<br>(0.060)  |                    |                     |                      |                    |
| ADDITHW                                                                                              | 0.041<br>(0.092)     | 0.387*<br>(0.205)   |                     |                     |                     |                      |                    |                     |                      |                    |
| DISS                                                                                                 | −0.077***<br>(0.027) | −0.112**<br>(0.041) | −0.012<br>(0.021)   |                     | 0.012<br>(0.038)    | −0.060***<br>(0.021) |                    | 0.047<br>(0.052)    |                      |                    |
| ATITMATH                                                                                             | 0.020<br>(0.018)     | 0.024<br>(0.028)    |                     |                     |                     |                      |                    |                     | 0.087<br>(0.066)     | 0.084<br>(0.100)   |
| hardHW_H                                                                                             |                      |                     |                     |                     |                     |                      |                    |                     | 0.046*<br>(0.066)    | 0.100<br>(0.409)   |
| $R^2$                                                                                                | 0.659                | 0.574               | 0.358               | 0.363               | 0.371               | 0.208                | 0.345              | 0.421               | 0.392                | 0.244              |
| Adj. $R^2$                                                                                           | 0.533                | 0.417               | 0.295               | 0.283               | 0.274               | 0.127                | 0.259              | 0.327               | 0.333                | 0.118              |
| $F$                                                                                                  | 5.23***              | 3.66**              | 5.71***             | 4.55***             | 3.84***             | 2.56*                | 4.00***            | 4.48***             | 6.62***              | 1.93               |

## Robustness: Autoregression (AR) and Endogeneity (2SLS)

Results remain consistent after introducing autoregressive terms (prior quiz or midterm score) and Two-Stage Least Squares instrumentation. (\* 0.10, \*\* 0.05, \*\*\* 0.01)

|      | Inducing | 1h0  | 1h0-r | 2h0 | 1h1 | 1h1-d | 1h1-i | 2h1  | 2h1-d | 2h1-i | 1h2 | 1h2-r | 2h2 | 2h2-r |
|------|----------|------|-------|-----|-----|-------|-------|------|-------|-------|-----|-------|-----|-------|
| AR   | ✓***     | ✓*** | ✓*    |     |     |       |       | ✓*** | ✓***  |       |     |       |     |       |
| 2SLS | ✓***     |      |       |     |     |       |       | ✓*** | ✓***  |       |     |       | NA  |       |

**Model legend:** 2h0 = Y<sub>2</sub> (Second Midterm), Hypothesis 0 -r = reduced -d = with DISS -i = with interaction term

## Concluding Summary

- ▶ **H0 — Supported:** DISS is a robust negative predictor (−7.6 to −11.2 pp per index point) across OLS, AR, and 2SLS. atitMATH is consistently insignificant.
- ▶ **H1 — Supported:** PROXI75 significantly improves scores and persists after controlling for dishonesty and in Y<sub>2</sub> when collusion was reduced — consistent with social learning. PROXI25 shows no effect.
- ▶ **H2 — Not supported:** No significant linear relationship found among attendance, homework orderliness, quiz scores, and perceived difficulty.

## Limitations & Future Directions

**Limitations:** Small single-cohort sample ( $N = 49$ ); homework not rubric-scored; honesty measures rely on observable proxies; prior maths grades unavailable.

**Future work:** Larger multi-cohort samples; rubric-based homework scoring; objective prior-ability measures; separated data collection and grading roles; panel and non-linear models.